

■ NON-NEWTONIAN FLUID — OOBLECK

What Is Oobleck? The Magic Goopy Science!

Planrs Academy | Kitchen Science Lab

■ Subject	Science Kitchen Experiments States of Matter
■ Duration	30 Minutes
■ Big Idea	Some substances are BOTH solid and liquid — and that is real science!
■ You Need	Corn flour (maize flour) • Water • Bowl • Food colour (optional)
■ Context	Indian Kitchen Science — CBSE / ICSE / State Board compatible
■ Prepared by	Planrs Academy — Curriculum Team

This lesson guide takes students from a mystery question — 'Is this magic or science?' — through hands-on making, testing, and discovering the extraordinary world of non-Newtonian fluids. Every experiment uses ingredients found in any Indian kitchen.

1 ■ The Hook — Is This Magic or Science?

Open the lesson with this electrifying question to spark curiosity before any explanation:

Have you ever seen something that looks like a liquid...
...but turns SOLID when you touch it? ■
What if you could actually WALK on goo without sinking?

■ Opening Hook Script (Teacher)

"Hold up a bowl. Pour a little of your pre-made Oobleck slowly — let it drip. Then suddenly SLAM your hand on the surface. Watch every student's face! Ask: What just happened? Accept ALL guesses. Wrong guesses are brave guesses!"

2 ■ Is This a Solid or a Liquid?

Understanding the Three States — Indian Kitchen Edition

■ SOLIDS	■ LIQUIDS	■ IN-BETWEEN	■ THINK ABOUT IT!
Keep their shape: • Jaggery (gur) • Chapati • Ice cube • Rock salt	Flow and pour: • Milk • Coconut oil • Lemon water • Lassi	Sometimes both: • Wet beach sand • Curd when stirred • Honey dripping	What if something is BOTH a solid AND a liquid at the SAME time? That's Oobleck! ■

■ Key Scientific Idea

- Most things are either solid (keeps its shape) or liquid (flows and pours).
- But some special substances switch between both depending on HOW you touch them!
- This makes Oobleck one of the most fascinating substances in everyday science.

3 ■■ Meet the Non-Newtonian Fluid!

■ What It Is	A non-Newtonian fluid is a special liquid that CHANGES how it behaves when force is applied. Normal (Newtonian) fluids like water always flow the same way. Non-Newtonian fluids do not follow these rules!
■ Fast Force	Touch it FAST → the tiny cornflour particles jam together → it acts like a SOLID! The faster you hit or press, the harder it becomes.
■ Slow Force	Touch it SLOWLY → particles have time to slide past each other → it flows like a LIQUID! Tilt the bowl gently and it drips like honey.
■ The Name	Named after Sir Isaac Newton, who described how normal fluids behave. Oobleck 'breaks Newton's rules' — hence NON-Newtonian!
■ Origin of 'Oobleck'	The name comes from 'Bartholomew and the Oobleck', a Dr. Seuss story (1949) about a king who wanted something new to fall from the sky — and a goopy green substance fell!

■ Story Time — Little Arjun's Kitchen Discovery

Little Arjun watches his Dadi make chapati in the kitchen. He notices something cool — when Dadi kneads the atta dough slowly, it feels smooth and soft like clay. But when she presses it fast with her palm, it pushes back hard! "Dadi, why does it do that?" he asks. Dadi smiles — that's just like our magic Oobleck, Arjun! Some things change their behaviour depending on how you touch them. That is not magic. That is SCIENCE!

4 ■ Only 2 Things from Your Kitchen!

Materials List — Indian Kitchen Edition

#	Material	Indian Name / Notes	Cost
1	Corn flour / Cornstarch	Maize flour (■■■■■ ■■■). The white powdery flour, NOT the coarse atta. Found in any kirana store or supermarket. Ask Mummy or Dadi to find it!	■20–40
2	Water	Plain tap water at room temperature. Do NOT use hot water — it changes the texture. Add slowly, bit by bit.	Free
3	Food colour (Optional)	Any liquid food colour — green works best visually! 1 drop is enough. Without colour, Oobleck is white/grey.	■10–20
4	A wide mixing bowl	Katori or large steel plate (thali) works perfectly. Wider = easier to mix and test!	At home
5	Your hands!	The best tool for feeling the magic. Oobleck is safe and non-toxic. Wash hands before and after.	Free

■ Teacher Prep Checklist

- Pre-make a batch of Oobleck before the session — ratio is 2:1 (flour:water).
- Test the consistency: it should drip slowly but resist sudden pokes.
- Have a spare batch ready — children love to experiment!
- Total cost per student: under ■30. Materials available at any kirana store.
- Cover the desk with newspaper or an old plastic sheet for easy cleanup.

5 ■ How to Make Oobleck — Step by Step

The Magic Ratio: 2 parts flour : 1 part water

STEP 1 Pour the Corn Flour

- Take a wide mixing bowl (katori or thali).
- Measure 1 cup (about 120g) of corn flour / maize flour.
- Pour it into the bowl. Notice how light and powdery it feels!
- TIP: Make sure your bowl is wide — you need space to mix and test.

STEP 2 Add Water Slowly

- Measure half a cup (about 60ml) of plain room-temperature water.
- Pour the water SLOWLY into the flour — a little bit at a time.
- Watch how it changes as you add each drop — this is the magical moment!
- Do NOT add all the water at once — the consistency changes gradually.

STEP 3 Add Colour (Optional)

- Add just 1 drop of food colour into the mixture.
- Green looks most like 'goopy science'. Pink, blue, or yellow all work!
- Stir gently — the colour will spread through the Oobleck.
- Fun fact: Oobleck is named after a green goo from a Dr. Seuss story!

STEP 4 Mix With Your Hands

- Stir slowly with your fingers. Feel the texture change as you mix.
- Keep mixing until it feels smooth and goopy — no dry lumps.
- The magic ratio is 2:1 — two parts flour to one part water.
- If too thick: add a few drops more water. If too runny: add a pinch of flour.

STEP 5 Test It Is Ready!

- Poke it fast — it should feel solid and resist your finger.
- Tilt the bowl — it should drip slowly like thick honey.
- Roll some between your palms quickly — it should form a ball!
- Stop rolling and hold still — the ball should melt back into liquid. ■

■ Safety Notes

- Corn flour is non-toxic and food-safe — completely safe for children.
- Do NOT eat Oobleck if food colour has been added.
- Keep Oobleck away from drains — it can clog pipes when dry.
- Dispose in a dustbin when done. It hardens when it dries.
- Wash hands thoroughly after handling.

6 ■ Four Tests — Let's Explore!

Run each test and observe what happens!

TEST 1: Slow Pour ■

Tilt your bowl and let Oobleck drip slowly. Watch it flow smoothly like honey or coconut oil. It acts just like a LIQUID when you are gentle with it!

TEST 2: Fast Poke ■

Poke the Oobleck really fast with your finger. Surprise! It feels hard like a solid. The faster you push, the harder it gets! It will not let your finger in easily.

WHAT HAPPENS? ■

When you move slowly, the tiny particles have time to slide past each other — just like people slowly walking through a crowd. That is why it flows! Slow = LIQUID.

TEST 3: Roll Into a Ball ■

Roll the Oobleck quickly between your palms. It forms a ball and holds its shape! It acts like a SOLID when you move fast. The faster you roll, the harder it feels!

TEST 4: Stop Rolling ■

Stop rolling and hold still. Watch the ball slowly MELT back into goopy liquid in your hands! It flows like water again — the particles unlock and slide past each other.

WHAT HAPPENS? ■

The tiny particles JAM together when pressure is applied quickly — like a crowd on a Mumbai local train at rush hour — nobody can move! That is the Oobleck locking up.

Student Observation Recording Table

Test	What I Did	What I Saw	Solid or Liquid?
1 — Slow Pour	Tilted the bowl slowly		
2 — Fast Poke	Poked it quickly		
3 — Roll a Ball	Rolled quickly between palms		
4 — Stop Still	Held still after rolling		

7 ■ The Science Behind Oobleck

Why Does It Do This? — Shear Thickening Explained Simply

■ Inside Oobleck	Tiny corn flour particles are packed tightly together inside water. They are so small you cannot see them — but you can FEEL them!
■ Fast Force → SOLID	When you press FAST, the particles have NO time to move out of the way. They JAM together and create a dense, hard surface. This is called SHEAR THICKENING — the substance gets thicker under sudden force.
■ Slow Force → LIQUID	When you move SLOW, the particles have plenty of time to slide past each other. They reorganise and flow — just like normal water would. This is why Oobleck drips and pours when you are gentle.
■ Mumbai Train Analogy	Think of the tiny particles like commuters on a Mumbai local train at rush hour! When everyone pushes at once (fast force) — nobody can move. When people move slowly one by one — they can slide past each other easily. That is EXACTLY what happens inside Oobleck!
■ The Science Word	This is called SHEAR THICKENING (or dilatant behaviour). It is a property of certain NON-NEWTONIAN fluids. Oobleck is one of the easiest and most dramatic examples to observe!

8 ■ Spot the Non-Newtonian — In Your Daily Life!

Indian Examples You Can Find Right Now at Home!

■ Wet Beach Sand

Walk fast on wet sand at Marina Beach, Goa, or Puri — it feels solid under your feet! But stand still and your feet slowly sink in like liquid. Classic non-Newtonian!

■ Idli Batter

Stir idli batter slowly — it flows easily. Stir it fast and it becomes thick and resists your spoon! Non-Newtonian right in your kitchen!

■ Ketchup Bottle

Shake the ketchup bottle hard to make it flow. This one works the OPPOSITE way — it gets thinner with force! That is called SHEAR THINNING. Interesting, right?

■ Atta Dough

Knead atta slowly — it feels smooth and soft. Press hard and fast — it pushes back! Just like Oobleck! Arjun's Dadi was doing non-Newtonian science all along!

■ Fevicol / Glue

Pull Fevicol slowly — it stretches like taffy. Pull it fast — it snaps and breaks! This shows how force speed changes the behaviour of sticky fluids.

■ Quicksand

Quicksand (and Oobleck!) both get HARDER when you struggle and push fast. The correct escape from quicksand is to move very SLOWLY — just like with Oobleck!

9 ■ What If You Tried This? — Extension Experiments

Wonder and Experiment More! Inquiry-Based Extensions

What If...	What You Need	What to Observe	Science Link
Add MORE colour drops?	Extra food colour	Does colour change the way Oobleck feels or behaves?	Viscosity vs colour additives
Put Oobleck on a speaker with LOUD music?	Speaker, phone	Watch the Oobleck 'dance' and form shapes! (Cymatics)	Sound waves + non-Newtonian fluids
Mix it with glitter?	Craft glitter	Does it still behave the same? Does glitter change anything?	Additives & suspension
Use SALT water instead of plain water?	Salt, water, flour	Does salt water change the texture or behaviour?	Dissolved solids & viscosity
Add MORE flour or MORE water?	More flour or water	What is too thick? What is too thin? Find the perfect ratio!	Concentration & shear thickening
Try ARROWROOT powder (araroot)?	Araroot powder	Does araroot make the same kind of Oobleck as corn flour?	Different starches & Non-Newtonian

■ Real-World Science Applications

- Bulletproof vests: non-Newtonian fluid panels harden on impact, protecting soldiers.
- Speed bumps: some experimental roads use non-Newtonian fluid — gentle cars pass, fast ones are slowed!
- Body armour for athletes: liquid armour in sports gear hardens on collision.
- Quicksand rescue: knowing it is non-Newtonian tells you to move slowly to escape.
- Medical research: blood is also non-Newtonian — it gets thinner as your heart pumps faster!

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■ Science Vocabulary Learned Today

Key Words — Understand, Pronounce, Use in a Sentence!

NON-NEWTONIAN FLUID <i>/Non-NEW-tonian FLOO-id/</i>	A liquid that changes its behaviour when force is applied — unlike normal liquids.
SOLID <i>/SOL-id/</i>	A state of matter that keeps its shape. Example: jaggery, ice cube.
LIQUID <i>/LIK-wid/</i>	A state of matter that flows and takes the shape of its container. Example: milk, water.
SHEAR THICKENING <i>/Sheer THICK-en-ing/</i>	Getting THICKER and harder when sudden force is applied. Oobleck does this!
VISCOSITY <i>/Vis-KOS-i-tee/</i>	How thick or thin a liquid is. Honey has high viscosity. Water has low viscosity.
PARTICLES <i>/PAR-ti-kuls/</i>	Tiny pieces that make up all matter. In Oobleck, corn flour particles jam together.
SUSPENSION <i>/Sus-PEN-shun/</i>	When particles are spread through a liquid but not dissolved. Oobleck is a suspension!
OBSERVATION <i>/Ob-zer-VAY-shun/</i>	Watching carefully and noticing details. The most important skill in science!

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■ Assessment — Exit Quiz

5 Questions — What Did You Discover?

Q #	Question	Type	Answer Key
Q1	Oobleck is made from ____ and ____.	Fill-in-blank	Corn flour (maize flour) and water
Q2	What happens when you POKE Oobleck very fast?	Multiple Choice: A. It flows B. It gets hard C. It disappears	B — It gets hard like a solid
Q3	TRUE or FALSE: Oobleck always behaves like a liquid.	True / False	FALSE — it changes based on force applied
Q4	Name ONE Indian food or substance that behaves like Oobleck.	Open-ended	Any of: atta dough, idli batter, wet beach sand, ketchup, Fevicol
Q5	Explain in one sentence why Oobleck becomes solid when hit fast.	Written	Tiny particles jam together when sudden force is applied, creating a solid surface.

Observation Rubric (Teacher Use)

Score	What It Looks Like
3 ■■■	Completed all 4 tests, recorded observations clearly, linked behaviour to solid/liquid, asked an extension question.
2 ■■	Completed at least 2 tests, made clear observations, could say 'fast=solid, slow=liquid'.
1 ■	Participated in experiment, made at least 1 observation, could name one ingredient.
0	Did not participate — provide hands-on support and re-demonstrate with teacher guidance.

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2

■ 3 Things You Learned Today

1

OOBLECK = CORN FLOUR + WATER | Just two simple ingredients from your kitchen create one of the most fascinating science demonstrations possible. Mix them in the ratio 2:1 and the magic begins!

2

IT IS A NON-NEWTONIAN FLUID | This special substance does NOT follow the normal rules of liquids. It changes how it behaves depending on HOW you touch it — breaking Isaac Newton's rules every single time!

3

PRESS FAST = SOLID | MOVE SLOW = LIQUID | When you apply sudden force, particles jam together and it becomes hard. When you move gently, particles slide past each other and it flows. That is shear thickening — and it is real science!

■ Great Job, Young Scientist! ■

You made something magical from your kitchen — and you understand WHY!

Keep asking WHY and WHAT IF — that is exactly what scientists do!

✓ The Complete Oobleck Discovery Summary

- ★ Oobleck = Corn flour + Water (2:1 ratio) ★ Non-Newtonian fluid — breaks Newton's rules
- ★ Fast force → particles JAM → SOLID behaviour ★ Slow force → particles SLIDE → LIQUID behaviour
- ★ Indian examples: atta dough, idli batter, wet beach sand, ketchup
- ★ Real-world use: bulletproof vests, liquid armour, medical research!